

KnoBAB

Fast Business Process Management with Temporal Reasoning

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- Procedural vs Declarative Temporal Models
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- Conformance Checking
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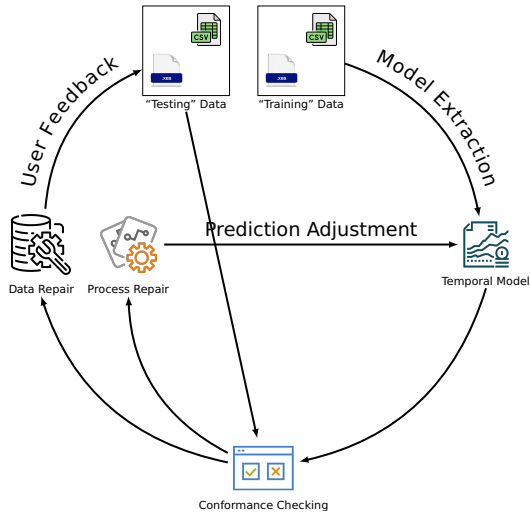
2 KnoBAB

- What is KnoBAB?
- Temporal Algebraic Operators (xtLTL_f)
- Conjunctive and Aggregation Queries
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Plan II

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- Temporal Model Mining and Conformance Checking
- Query Plan Parallelisation
- Minimal Model Description Length
- Future Works and Research Collaborations
- Transparency and Result Replicability

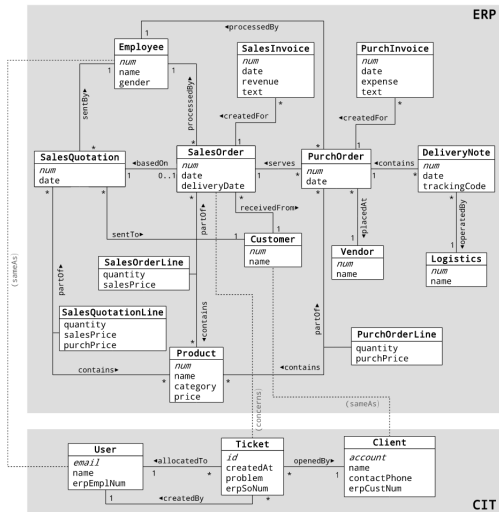
Business Process Management Ecosystem



A complete Business Process Management Ecosystem allows you to:

- Extract **temporal models** from the logs.
- Check whether the model (or the new data) **conforms** to the given temporal model.
- If not, then repair either the model (**continual learning**) or the data (**User Feedback**).

Running Example



FoodBroker (1/3)

A fictive company trading food offers a brokerage service between producers (vendors) and retailers (customers). Process-related data is recorded in an enterprise resource planning (ERP) system and a customer issue tracking (CIT) system.

The *Logs* are the descriptions of the *traces* (i.e., transactions) stored in your ERP system. Logs can represent both Activity Labels as well as *data payloads* with literal or numerical values.

Logs can be serialised in different formats:

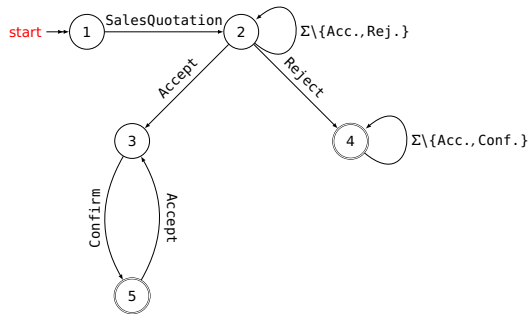
- CSV
- XES (XML: *de facto standard*)
- Human-Readable Format (HRF)

FoodBroker (2/3)

Each trace starts with a *vendor* sending a SalesQuotation to a *customer*. If an *offer* is accepted and the OrderConfirmed, then the *item* is Scheduled for delivery. When ready, a *logistic operator* Collects it and eventually Delivers it.

Procedural (*left*) vs Declarative (*right*) Temporal Models (1/3)

Temporal Models describe the expected behaviour of the system. A simplified description of the previous model can be determined as follows:



- Init(SalesQuotation)
- ExclChoice(Accept,Reject)
- NotSuccession(Reject,Confirm)
- ChainResponse(Accept,Confirm)

We will later discuss how such models might be extracted...

Procedural (*left*) vs Declarative (*right*) Temporal Models (2/3)

Despite the current hype on graphs, procedural model come in this field with several shortcoming:

- It can be proven that a graph might have an exponential size if compared to the declarative model.
 - ⇒ Graph models are harder to visualize.
- A clause-based formulation might lead to further algorithmic optimizations.
- Basic changes in the declarative model entails convoluted procedural edits.
- Procedural models cannot adequately express correlation conditions between events happening at remote times (unless we exhaustively enumerate the conditions satisfying such correlation).
- Declarative models specify temporal constraints among activities without strictly enforcing their order of completion.

Procedural (*left*) vs Declarative (*right*) Temporal Models (3/3)

Declarative languages can be expressed in terms of algebraic or logical operators:

- A clause-based formulation might lead to further algorithmic optimizations.
- Declare is not the only possible declarative language, as many possible declarative languages can be identified via specific temporal patterns of interest.

Still, current business process management systems support only one declarative language at a time...

Use Cases: e-Health (1/2)

INPUT DATA (HUMAN READABLE)	
Log	
①	Trace Payload { loc_po = "LN", p_id = "001A" }
	Referral { CA_15-3 = 69 }
	Mastectomy { CA_15-3 = 69, biopsy = true }
	FollowUp { CA_15-3 = 10 }
②	Trace Payload { loc_po = "NE", p_id = "002A" }
	Referral { CA_15-3 = 20 }
③	Trace Payload { loc_po = "YO", p_id = "003A" }
	Referral { CA_15-3 = 61 }
	Lumpectomy { CA_15-3 = 61, biopsy = true }
	FollowUp { CA_15-3 = 55 }

Max-SAT Query	
Declare Model (M)	
Ⓐ	Response (Referral, CA_15-3 >= 23.5, FollowUp, CA_15-3 < 23.5) where Referral.CA_15-3 > FollowUp.CA_15-3
Ⓑ	Succession (Referral, CA_15-3 >= 23.5, FollowUp, CA_15-3 < 23.5) where Referral.CA_15-3 > FollowUp.CA_15-3
Ⓒ	Choice (Mastectomy, CA_15-3 >= 50 && biopsy = true, Lumpectomy, CA_15-3 >= 50 && biopsy = true)

- *openEHR*: A medical process describes clinical-related procedures as well as organisational management ones (e.g., registration, admission, and discharge).
- A patient's clinical course can be then collected in a log.
- Temporal models can detect discrepancies between clinical guidelines, expressed as temporal models, and some collected traces.

Use Cases: e-Health (2/2)

Log:

- ① A cancer patient with a successful mastectomy
- ② A healthy patient
- ③ An unsuccessful lumpectomy suggesting that the patient might still have some cancerous cells.

Model:

- Ⓐ-Ⓑ Any successful treatment should decrease the CA-13.5 levels, which should be below the guard level.
- Ⓒ Two possible surgical operations for breast tumours are mastectomy or lumpectomy if the biopsy is positive and the CA-13.5 is way above the guard level.

INPUT DATA (HUMAN READABLE)	
Log	
①	Trace Patient (loc, pos = "LBN", p, id = "000A") Referal (CA_13.5 = 68) Mastectomy (CA_13.5 = 68, biopsy = true) FollowUp (CA_13.5 = 18)
②	Trace Patient (loc, pos = "TMR", p, id = "000B") Referal (CA_13.5 = 20)
③	Trace Patient (loc, pos = "TDP", p, id = "000A") Referal (CA_13.5 = 61) Lumpectomy (CA_13.5 = 81, biopsy = true) FollowUp (CA_13.5 = 58)

Max-SAT Query	
Declare Model (M)	
Ⓐ	Response: Referal: CA_13.5 >= 23.5, FollowUp: CA_13.5 < 23.52 where Referal: CA_13.5 > FollowUp: CA_13.5
Ⓑ	Sentence: Referal: CA_13.5 >= 23.5, FollowUp: CA_13.5 < 23.52 where Referal: CA_13.5 > FollowUp: CA_13.5
Ⓒ	Choice: Mastectomy: CA_13.5 >= 58 && biopsy = true, Lumpectomy: CA_13.5 >= 50 && biopsy = true

Given a log and a temporal model, we want to assess the degree of compliance of a log to a temporal model, or vice-versa:

- By exploiting a declarative model, we can directly assess how many temporal constraints were satisfied (*MaxSAT*) or evaluate *Confidence* or *Support* of the given model rules.
- We can also filter the traces completely satisfying the model.

FoodBroker (3/3)

Non-conformant traces (called **deviant**) either do not reflect the company's rules or lead to retailers' complaints.

A late delivery complaint can occur only if the *date* the product is Delivered is greater than the *agreed time* to receive it as registered the OrderConfirmation:

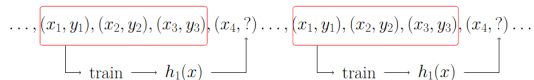
`Response(OrderConfirmation,Delivered)`
where *agreed_time* $\leq$ *date*.

Use Case: Cyber-Security (1/3)

Intrusion detection can audit an environment for logging sources of potential flaws that can be remedied and fixed later.

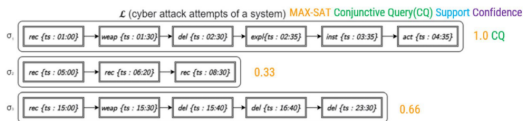
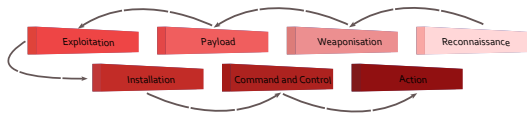
- **anomaly-based** approaches raise an alarm if the observed behaviour differs significantly from the expected one
- **signature-based** approaches check whether attack patterns might be recognised from the environment.

Signature-based approaches often mitigate the high false-alarm rates of anomaly-based ones.



Lifelong Learning on stream data from finite temporal windows

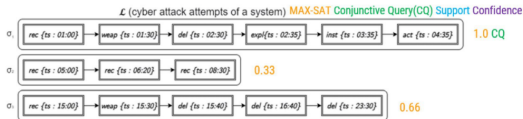
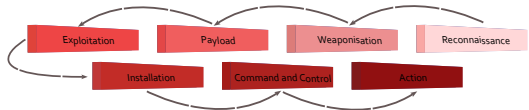
Use Case: Cyber-Security (2/3)



M		
	Clause	LTL _i
Ⓐ	Response(rec, true, weap, true)	$\Box(\text{rec} \Rightarrow \Diamond(\text{weap} \wedge \text{true}))$ 2 / 3 = 0.66, 2 / 3 = 0.66
Ⓑ	Absence(iot_sh, true, 1)	$\neg(\text{Exists}(\text{iot_sh} \wedge \text{true}, 1))$ 3 / 3 = 1.0, 3 / 3 = 1.0
Ⓒ	Choice(comm, true, act, true)	$\Diamond(\text{comm} \wedge \text{true}) \vee \Diamond(\text{act} \wedge \text{true})$ 1 / 3 = 0.33, 1 / 1 = 1.0

- *Reconnaissance*: the attacker mainly collects information regarding the system's vulnerabilities.
- *Weaponisation*: construction of the malware/attack.
- *Payload*: disguising the malware.
- *Exploitation*: dodging the defence mechanism through the disguise.
- *Installation*: the attack is installed in the system.
- *Command*: synch with the remote attacker.
- *Attack*

Use Case: Cyber-Security (3/3)



M			
	Clause	LTL _f	
Ⓐ	Response(rec, true, weap, true)	$\square(\text{rec} \Rightarrow \diamond(\text{weap} \wedge \text{true}))$	$2/3 = 0.66, 2/3 = 0.66$
Ⓑ	Absence(iot_sh, true, 1)	$\neg(\text{Exists}(\text{iot_sh} \wedge \text{true}, 1))$	$3/3 = 1.0, 3/3 = 1.0$
Ⓒ	Choice(comm, true, act, true)	$\diamond(\text{comm} \wedge \text{true}) \vee \diamond(\text{act} \wedge \text{true})$	$1/3 = 0.33, 1/1 = 1.0$

- The only trace having a completely successful attack is σ_1 : this might be returned by a **Conjunctive Query**.
- *Support* determines that Ⓒ occurred far way less than Ⓑ despite their *Confidence*.
- *Support* metrics can be also used to extract frequently-occurring event patterns occurring in the data.

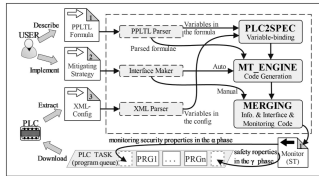
Model Mining

- A model mining approach will use *Support* and *Confidence* to extract frequently-occurring patterns from the training data.
- Model mining is still a required step before performing declarative model learning (*propositionalisation*).

Model Learning

- The life-learning approach with sliding windows is model- and data-representation independent.
- Upon trace labelling, we can use learning approaches for providing minimal model descriptions distinguishing differently-classified traces.

Use Cases: Industry 4.0 (1/4)



Linear Time Logic in action over PLCs.

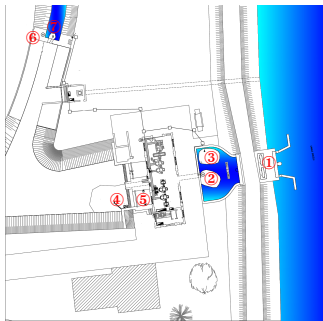


- **Smart factories** enable the collection and analysis of data through advanced sensors and embedded software for better decision-making.
- Run-time verification can be deployed as a permanent testing condition on the environment, thus reducing **model-checking** to **conformance checking**.
- At each scan cycle, **Programmable Logic Controllers** controllers perceive through sensors the status change of the environment (e.g., variations of temperature and pressure). This information is then fed to the internal logic, which, on the other hand, might decide to intervene directly in the environment by sending signals to some actuators.

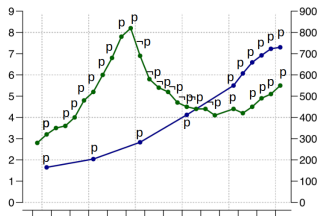
Land Reclamation Consortium

A *land reclamation consortium* is a public law body which takes care of the maintenance of public reclamation works, such as ensuring that sunken areas are not flooded by heavy rains, and upkeep the existing pump and irrigation plants.

To replenish the *Bianco canal* upstream ④, the water is collected from the river Po via the *Boicelli canal* ① via an irrigation sluice ②. The level of the Bianco canal downstream is regulated via a sluice gate ⑥. Sensors on both ④ the Boicelli and the Bianco canal (④ upstream ⑦ downstream) tracks different water depth levels; we also monitor the power consumption of each motor governing the sluices' opening.



Use Cases: Industry 4.0 a) Temporal Time Series Patterns (3/4)



Log Representation

We can *propositionalise* temporal time series as follows:

- Each data point where the water level increases is associated to an activity label p
- ...and to a different label otherwise (e.g., $\neg p$).

Differently from current literature, we can now also embed data values directly from the data point!

Defining a new declarative language

As all the major declarative languages support indirectly Linear Temporal Logic for Finite traces, we can express temporal variations of those through novel declarative templates. KnoBAB also allows to express data correlations!

PLTL formula	Line type	Data trends	Form
$\Box p$		Increase rapidly	V
$\neg p \rightarrow \Box p$		Increase slowly	IV
$p \rightarrow \Box \neg p \rightarrow \Box p$		Increase slowly	IV
$p \rightarrow \Box p \rightarrow \Box \neg p \rightarrow \Box p$		Increase slowly	IV
$\Box p \rightarrow \Diamond \neg p$		Increase slowly	IV
$(\neg p \rightarrow \Box(p \wedge \neg p)) \wedge (p \rightarrow \Box p)$		High volatility	III
$(\neg p \rightarrow \Box p) \wedge (p \rightarrow \Box \neg p)$		High volatility	III
$(p \rightarrow \Box(p \wedge \neg p)) \wedge (\neg p \rightarrow \Box p)$		High volatility	III
$(\neg p \rightarrow \Box(p \wedge \neg p)) \wedge (p \rightarrow \Box \neg p)$		High volatility	III
$(p \rightarrow \Box \neg p) \wedge (\neg p \rightarrow \Box p)$		High volatility	III
$(p \rightarrow \Box(p \wedge \neg p)) \wedge (\neg p \rightarrow \Box \neg p)$		High volatility	III

Use Cases: Industry 4.0 **b)** Actuators Breakdown Control (4/4)

If we are in the summer and we water level downstream is below the minimum threshold min_1 , then we increase the sluice opening for ϑ_1 inches...

ChainResponse(WaterSensing, $downstream < min_1$,
Sluice, $d_in = \vartheta \wedge d_open = \mathbf{true}$)

...and that should be still open until the downstream level reaches the level.

WeakUntil(WaterSensing, $downstream \geq min_1 \wedge downstream \leq max_1$,
Sluice, $d_in = \vartheta_1 \wedge d_open = \mathbf{true}$)

If, as a consequence of this, the upstream level is below the desired threshold, we increase the irrigation sluice opening by ϑ_2 inches.

ChainResponse(WaterSensing, $upstream < min_2 \wedge d_open = \mathbf{true}$,
Sluice, $irr_in = \vartheta_2 \wedge irr_open = \mathbf{true}$)

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- Customisability
- Column-Store Data Representation

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What is KnoBAB?

- My specific research interest is on bridging the gap between theoretical research and practical use case scenarios.
- Despite most of the previous concepts are quite known in current literature, they might have either of the following shortcomings:
 - 1 Real-World comes with a huge amount of data coming in a **semi-structured** representation.
 - 2 Do not reap big data algorithms for carrying out computations efficiently
 - 3 Most of the time, the users are forced to “reduce” a data-aware domain to a dataless one to carry out business process mining operations (e.g., trace repairs or alignments).
 - 4 KnoBAB shows that it is possible to import a lot of good practices from the data science field to the benefit of a larger community.

I. Temporal Algebraic Operators (xtLTL_f)

Walking on the footsteps of Declare, we might assume to have activation and target condition to be tested. Differently from LTL_f, we also need to explicitly express data correlation conditions.



"Logic is the anatomy of thought"
John Locke (1632-1704)



Edgar Codd (1923-2003)

I. Temporal Algebraic Operators (xtLTL_f)

xtLTL_f: Syntax

Given optional A/T fields, an xtLTL_f expression is defined as follows:

$$\rho ::= \mu | \nu | \beta | \delta$$

$$\mu ::= \text{Activity}_{A/T}^{\mathcal{L}, \tau}(a) | \text{Compound}_{A/T}^{\mathcal{L}, \tau}(a, lower \leq \kappa \leq upper) | \text{First}_A^{\mathcal{L}, \tau} | \text{Last}_A^{\mathcal{L}, \tau} | \text{Init}^{\mathcal{L}}(a) | \text{Ends}^{\mathcal{L}}(a)$$

$$\nu ::= \text{Exists}_n(\rho) | \text{Absence}_n(\rho) | \text{Init}(\rho) | \text{Ends}(\rho) | \text{Next}^{\tau}(\rho) | \text{Globally}^{\tau?}(\rho) | \text{Future}^{\tau?}(\rho) | \text{Not}^{\tau?}(\rho)$$

$$\beta ::= \text{Until}_{\Theta}^{\tau?}(\rho_1, \rho_2) | \text{And}_{\Theta}^{\tau?}(\rho_1, \rho_2) | \text{Or}_{\Theta}^{\tau?}(\rho_1, \rho_2)$$

$$\delta ::= \text{AndFuture}_{\Theta}^{\tau}(\rho_1, \rho_2) | \text{AndGlobally}_{\Theta}^{\tau}(\rho_1, \rho_2) | \text{AndNextGlobally}_{\Theta}^{\tau}(\rho_1, \rho_2)$$

xtLTL_f: Output Semantics

All expressions ρ return an ordered sequence of nested records $\langle i, j, L \rangle$, entailing that the j -th event of the i -th trace satisfies the conditions of ρ . L lists all the data activation, target, and Θ correlation conditions being witnessed.

I. Temporal Algebraic Operators (xtLTL_f)

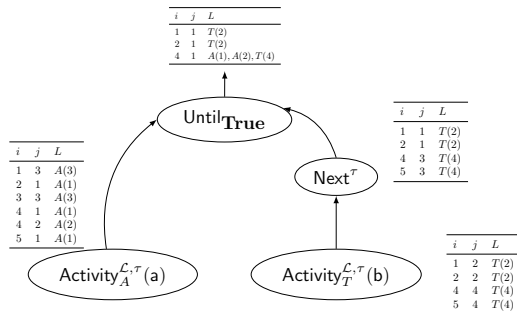
Tutorial

Let's suppose to evaluate $a\mathcal{U} \circ (b)$ over the following log:

$$\mathcal{L} = \{\text{dbadd}, \text{ab}, \text{cda}, \text{aadb}, \text{addb}\}$$

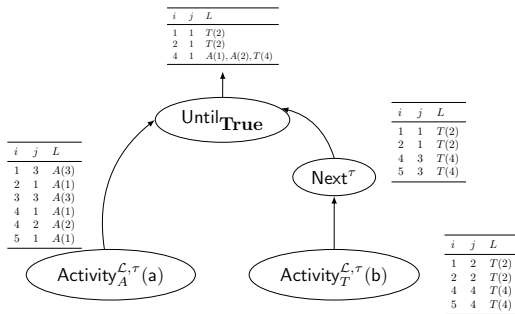
This can be expressed as

$$\text{Until}_{\text{True}}(\text{Activity}_A^{\mathcal{L}, \tau}(a), \text{Next}^{\tau}(\text{Activity}_T^{\mathcal{L}, \tau}(b)))$$



I. Temporal Algebraic Operators (xtLTL_f)

	ActivityId	Trace	Event	Prev	Next
1	d	1	1	⊥	8
2	d	1	4	12	3
3	d	1	5	2	⊥
4	d	3	2	18	14
5	d	4	3	16	10
6	d	5	2	17	7
7	d	5	3	6	11
8	b	1	2	1	12
9	b	2	2	13	⊥
10	b	4	4	5	⊥
11	b	5	4	7	⊥
12	a	1	3	8	2
13	a	2	1	⊥	9
14	a	3	3	4	⊥
15	a	4	1	⊥	16
16	a	4	2	15	5
17	a	5	1	⊥	6
18	c	3	1	⊥	4



II. Conjunctive and Aggregation Queries

Without the need of changing the operators' semantics or the results being returned, the same operators can express over the same “model” all of the following queries, by simply changing the root node of the query plan:

$$\text{CONJUNCTIVEQUERY}(\rho_1, \dots, \rho_n) = \text{And}_{\text{True}}(\rho_1, \dots, \text{And}_{\text{True}}(\rho_{n-1}, \rho_n))$$

$$\text{Max-SAT}(\rho_1, \dots, \rho_n) = \left(\frac{|\{l | \exists j, L. \langle i, j, L \rangle \in \rho_l\}|}{|\mathcal{M}|} \right)_{\sigma^i \in \mathcal{L}}$$

$$\text{CONFIDENCE}(\rho_1, \dots, \rho_n) = \left(\frac{|\{i | \exists j, L. \langle i, j, L \rangle \in \rho_l\}|}{|\text{ActLeaves}(\rho_l)|} \right)_{c_l \in \mathcal{M}}$$

$$\text{SUPPORT}(\rho_1, \dots, \rho_n) = \left(\frac{|\{i | \exists j, L. \langle i, j, L \rangle \in \rho_l\}|}{|\mathcal{L}|} \right)_{c_l \in \mathcal{M}}$$

III. Customisability

```
load TAB "/home/giacomo/test.tab" as "tab";

display ACTTABLE for "tab";

auto-timed queryplan "aaai23" {
  template "Example" args 2 := (EXISTS 1 t #1 activation) U
                                (NEXT EXISTS 1 t #2 target)
};

model-check
  declare "Example" ("a", true, "b", true)
using "TraceIntersection" over "tab"
plan "aaai23" with operators "Hybrid"
display query-plan;
```

```
/* Loading Log */

/* Dump the internal representation as CSV */

/* Setting the semantics for declarative op. */

/* Conformance Checking */
/* Manually providing the model */
/* Conjunctive query + Log name */
/* Semantics to adopt, type of algorithms */
/* Provide the DAG query plan */
```

- By supporting xtLTL_f instead of declarative clauses, our system allows the definition of customary templates (**queryplans**) that can be loaded at runtime.
- The running example could be run in KnoBAB (knobab_server) with the commands given above on a console (knobab_client).

IV. Interoperability

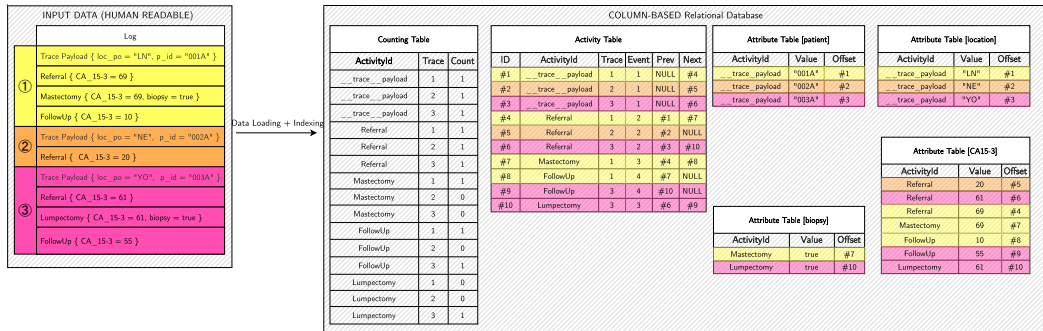


```
import pandas as pd
from io import StringIO

# Setting up the client
cli = uk_knobab_kinj_KINJ("localhost", 8795)
# Generating a loading data query
q = uk_knobab_kinj_QueryCompiler.LoadDataQuery(uk_knobab_kinj_LogFormat.TAB,
                                                "/home/giacomo/test.tab", "tab",
                                                False, False, False)
# Sending the data loading request to the server
cli.PerformQuery(q)
# Asking to dump the ActivityTable as a CSV string
q = uk_knobab_kinj_QueryCompiler.DisplayData(uk_knobab_kinj_Displays.Display(uk_knobab_kinj_DisplayTable.ACT_TABLE, "tab"))
result = cli.PerformQuery(q)
# Loading the CSV as a Pandas Dataframe
pd.read_csv(StringIO(result.message))
```

- KnobAB can be queried as a restful HTTP server (knobab_server) and has API (KINJ) supporting major programming language by transpiling code in Haxe!
- The above example shows how to perform a query and getting answers in Python by connecting to the KnobAB server.

V. Column-Store Data Representation

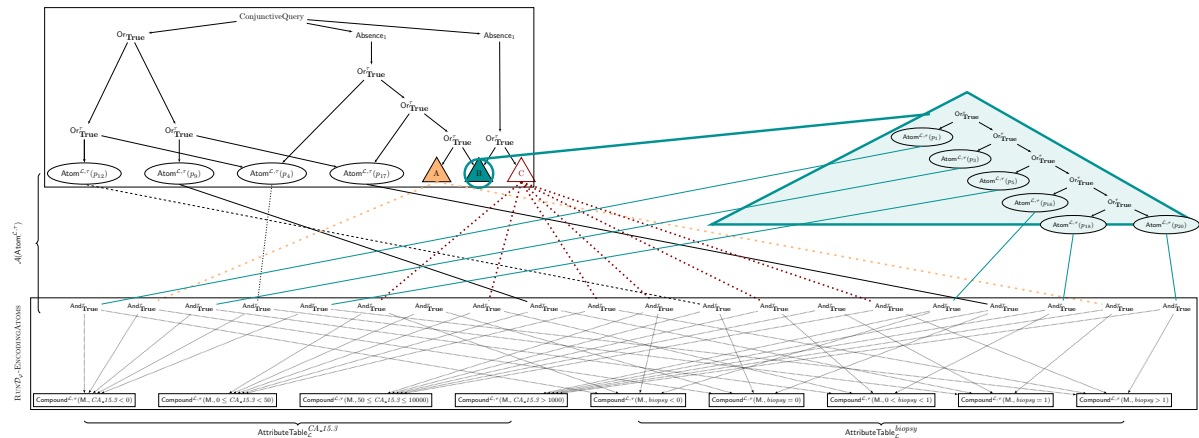


Supports different input data formats:

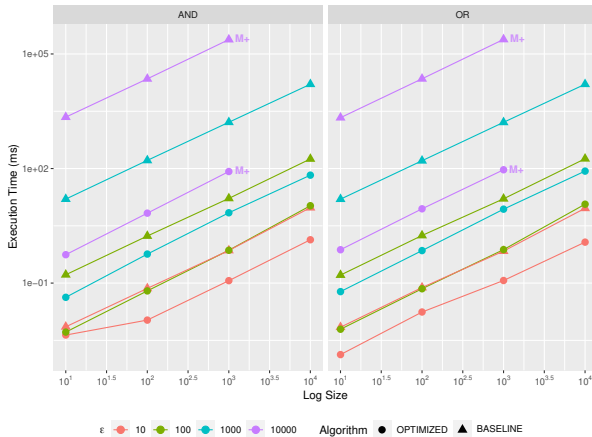
- Human Readable Format (BPM'21)
- XES
- Tab Separated Format

KnoBAB provides a Column-Base storage entailing minimal table representation when possible and providing a table decomposition of relationships otherwise. This also addresses log “data sparsity”...

VI. Query Plan Optimisation

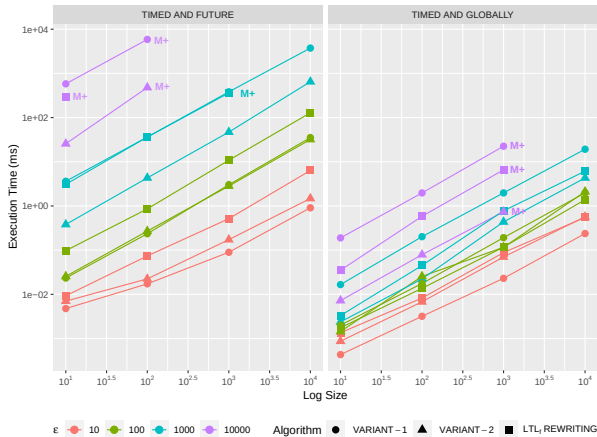


VI. Query Plan Optimisation



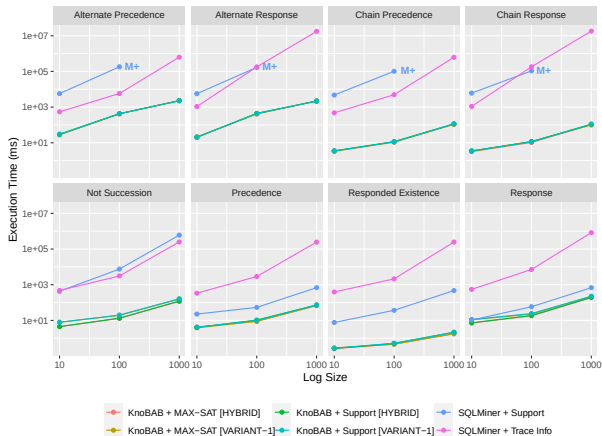
1. For untimed And and Or, we can provide different implementations, one being more memory efficient (Out of Primary Memory) and faster than the other.

VI. Query Plan Optimisation



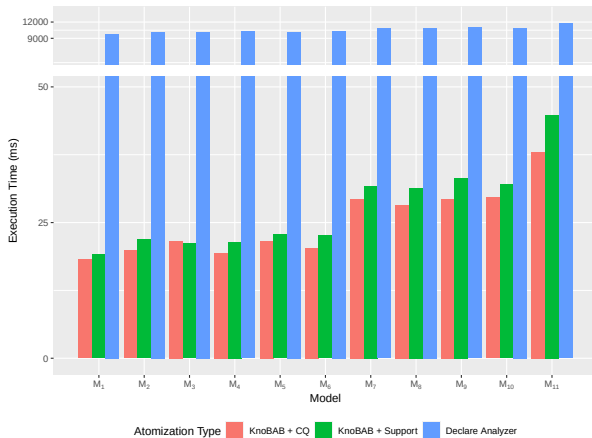
2. As customary in the database world, by defining *ad hoc* operators subsuming the evaluation of recurrent sub-expressions, we can design better algorithms!

VII. Temporal Model Mining...



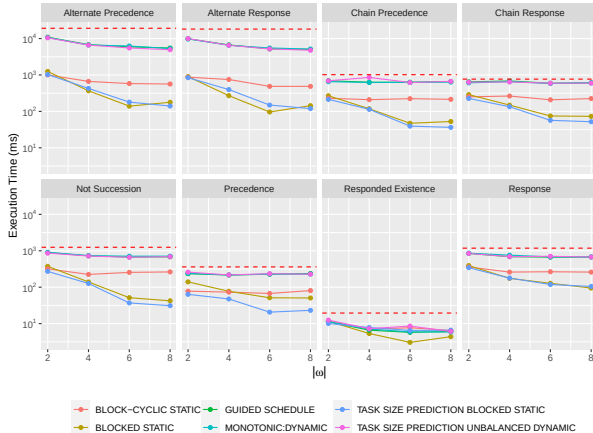
SQLMiner. If compared to other SQL-driven mining algorithms (where a given set of declarative clauses to be tested is previously generated through *propositionalisation*), our results show that our solution outperforms similar querying tasks on PostgreSQL.

VII. ...and Conformance Checking



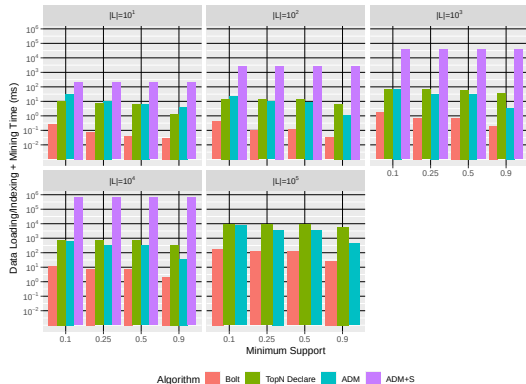
Declare Analyzer. Our solution also outperforms ad-hoc Java implementations for computing data-aware conformance checking solutions.

VIII. Query Plan Parallelisation



The DAG query plan allows running a *topological sort*. By “*layering*” the query plan, we can schedule which operators might be parallelised. Different scheduling policies give different speed-ups exploited by reducing page faults!

IX. Minimal Model Description Length



Log Gen. Model, $\mathcal{M}^*$	Bolt	TopN	ADM	ADM+S
ChainPrecedence(b, c)	✓	✗	✓(+Choice, CoExist.)	✓(+Choice, CoExist.)
ChainResponse(d, e)	✗	✗	✓(+Choice, CoExist.)	✓(+Choice, CoExist.)
Choice(f, g)	✓	✓(+Choice(g, f))	✓(+Choice(g, f))	✓(+Choice(g, f))
ExclChoice(h, a)	✓	✗	✗	✓(+ ExclChoice(a, h))
Exists(b, 1)	✓	✗	✗	✓
Init(e)	✓	✓	✓	✓
Precedence(c, b)	✓	✗	✓(+Choice, RespEx., CoEx.)	✓(+Choice, RespEx.)
RespExistence(e, f)	Response(e, f)	✓	✓	✓
RespExistence(h, i)	CoExistence(h, i)	✗	✓(+ CoExistence(h, i))	✓
RespExistence(i, h)	CoExistence(h, i)	✗	✓(+ CoExistence(i, h))	✓
Response(e, f)	✓	✓(+ RespExistence(e, f))	✓(+ RespExistence(e, f))	✓(+ RespExistence(e, f))
$ \mathcal{M}_\theta $	197	112	548	684
$ \mathcal{M}_{\geq 0.999} $	46	35	119	127
Mining time (μ)	$1.73 \cdot 10^0$ ms	$5.17 \cdot 10^1$ ms	$1.40 \cdot 10^2$ ms	$3.29 \cdot 10^3$ ms

- Our latest work shows that it is possible to obtain compact temporal models without additional learning tasks, by only enforcing non-zero confidence and traversing the lattice of the declarative patterns.
- Our solution also outperforms previous solutions, as well as renditions of previous algorithms on KnoBAB.

- It is possible to use KnoBAB as a library for re-implementing existing algorithms (ADM).

Future Works and Research Collaborations

- Fully support data-aware model mining and learning.
- Fully supporting secondary-memory storage for proper big-data representations.
- Supporting different data storages and representations for logs, such as graphs!
- Fully supporting trace and model repairs through machine learning algorithms.
- Distributed computation through efficient HPC programming standards (OpenMPI).
- GUI for providing visual information of the internals.

X. Transparency and Result Replicability

- The codebase is associated to test sets run with **googletest**.
- The papers' datasets are released on the **Open Science Framework**.
- We exploit **GitHub Workflows** for guaranteeing that the released code compiles and that all the tests are run satisfactorily.



googletest
Google C++ Testing Framework



<https://github.com/datagram-db/knobab>